

Technology Stack

Date	1 October 2026
Team ID	6ab501466f290be8a6b4f5d3
Project Name	ComicCraft – AI Comic Story Creator
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit (Python)	Lets us build a simple web interface with a prompt box, style selectors, image previews and a download button using only Python, without separate HTML or JavaScript.
2	Backend / Server-Side	Python	One language for the whole project. Separate modules handle story generation, image generation, layout and PDF export, which keeps the code easy to maintain.
3	Database / Data Storage	No database (Streamlit session state and temporary files)	The app does not need accounts or saved records. The story and generated panels are held for the user's session and the final comic is exported as a PDF.
4	Cloud / Hosting / Deployment	Runs locally with Streamlit (can be deployed on Streamlit Community Cloud)	Simple to run and demonstrate with a single command, with a free hosting option if the app needs to be shared online.
5	Version Control & CI/CD	Git and GitHub	Lets all team members work together, keeps a history of changes and is used to submit the project.
6	Third-Party APIs / Other Tools	Google Gemini API (text and image models), Pillow, FPDF2, python-dotenv	Gemini text model writes the story and panel script, and Gemini image model creates the panel pictures. Pillow arranges the panels into pages, FPDF2 exports the comic as a PDF, and python-dotenv keeps the API key secure.